

# Job control

Long builds and simulations should not lock your terminal forever

# Background, stop, and resume

- An ampersand at the end of a command starts it in the background, you get the prompt back
- Jobs lists this shell's jobs
- Control-Z suspends the foreground job; it is stopped, not killed
- Bg resumes a stopped job in the background
- Redirect output to a log, then tail the log while the job runs

# Browser lab

Digital Design and Verification Platform

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Quiz: jobsRun jobsFG job idQuiz: &Quiz: Ctrl+ZCtrl+Z makebg resumefg bringsleep &jobs -lQuiz: fg vs bgQuiz: not killCtrl+C fgQuiz: %1gtkwave bgRound tripQuiz: SIGTSTPmake PIDBusy fgQuiz: +TimelineQuiz: jobs vs ps

Lab terminal

Foreground: [1] PID 3101 · make sim

Ctrl+Z suspends · Ctrl+C kills · shell waits until this job finishes or stops.

Starter: [1] make sim (fg) · [2] gtwave (bg) – try Ctrl+Z then bg %1

lab\$ jobs · fg %1 · bg %1 · sleep 60 & · help

jobsjobs -lCtrl+Zbg %1fg %1sleep 60 &Ctrl+Cfg %2

jobs table

+ = current job - = previous fg / bg / Stopped

| JOB  | PID  | STATUS       | COMMAND          |
|------|------|--------------|------------------|
| [1]+ | 3101 | Running (fg) | make sim         |
| [2]- | 3102 | Running &    | gtkwave wave.vcd |

State flow

fg RunningCtrl+ZStoppedbgbg Runningfgfg Running

Timeline

Job control lab

# Real shell practice

```
wsl - module10-job-control-lab/examples/background

# Track A - real Unix
# real shell session (Track A)

$ set -m

$ chmod u+x slow_output.sh

$ ./slow_output.sh > bg.log 2>&1 &

$ jobs
[1]+  Running                  ./slow_output.sh > bg.log 2>&1 &

$ sleep 2

$ tail -5 bg.log
[03:12:42] Line 1
[03:12:43] Line 2

$ kill $(jobs -p); wait 2>/dev/null || true

$ jobs
```

## Real shell practice — try these

```
# chmod u+x slow_output.sh — make the demo script executable
chmod u+x slow_output.sh
```

```
# ./slow_output.sh > bg.log 2>&1 & — run in background; stdout/stderr
to
a log
./slow_output.sh > bg.log 2>&1 &
```

```
# jobs — list this shell's background jobs
jobs
```

```
# sleep 2 — give the script a moment to write lines
sleep 2
```

```
# tail -5 bg.log — show the latest log lines while the job runs
tail -5 bg.log
```

# Pitfalls to watch

- Control-Z stops
- Jobs only lists jobs for this shell, not every process on the machine
- Closing the terminal can hang or kill background jobs unless you plan for that
- And remember

# Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell, practice ampersand, jobs, and tailing a log
- When you are ready, take the short quiz, then continue to pipes, redirection, and xargs